Are pause-segmented chunks cognitive units? Evidence from CogARC Experiment 2 drawing trajectories

Hypothesis

Throughout our analyses of CogARC drawing behavior we have treated *chunks*—groups of consecutive cell edits separated by pauses longer than $\max(2 \cdot \tilde{r}_{\text{subject}}, 500\,\text{ms})$, where $\tilde{r}$ is a subject's median inter-edit reaction time—as the participant's own cognitive units. That is, our working assumption was that a chunk corresponds to "draw one thing, pause to decide, then draw the next thing." This assumption is load-bearing: it underwrites claims such as *early chunks predict trajectory outcome*, *chunk features distinguish motor from cognitive errors*, and our GNN's imitation target for step-by-step drawing. If chunks were instead arbitrary time slices imposed by the analyst, those claims would have no psychological content.

We therefore test directly:

> *Pause-segmented chunks reflect participants' own segmentation of the task—they are cognitive units, not analytical artifacts.*

If the hypothesis holds, (i) inter-edit reaction times should separate into a fast *intra-chunk* (motor) regime and a slow *inter-chunk* (planning) regime; (ii) real chunks should be *structurally coherent*—a chunk should tend to contain one color and form one connected spatial region—to a degree that does not survive null re-segmentations; and (iii) different subjects should *agree* on the task's partition, since if chunking were idiosyncratic the partitions would be effectively random across subjects.

Methods

We analyzed 399,684 chunks across 75 tasks and 233 Experiment-2 subjects from the CogARC release.[1] For every trajectory we computed chunks three ways:

Real. Pause-segmented with the standard threshold.

Null-RT. Inter-edit RTs shuffled within the trajectory, then re-segmented with the same pause rule. Preserves the RT distribution; destroys pause-ordering.

Null-Cut. Random cut points uniformly placed in the edit sequence, with the same number of chunks as Real. Preserves chunk count; ignores pauses entirely.

For each chunk we computed three structural-coherence features, all of which should be high for genuine cognitive units and close to baseline for the nulls:

[1] Submission sequences parsed via `human_chunking.identify_chunks`; Success-component targets from the task's common solutions.

- **color_homogeneity**: fraction of the chunk's edits that use the chunk's dominant color.
- **is_connected**: whether the chunk's cells form a single 4-connected component.
- **success_iou_best**: IoU between the chunk's cells and the best-matching Success-grid connected component.

For the cross-subject agreement test we computed the Adjusted Rand Index (ARI) between random subject pairs (50 pairs per task, 3,750 pairs total per condition), restricting each pair to cells painted by *both* subjects. Finally we tagged every inter-edit gap as intra-chunk or inter-chunk according to that subject's threshold, yielding two empirical RT distributions.

Results

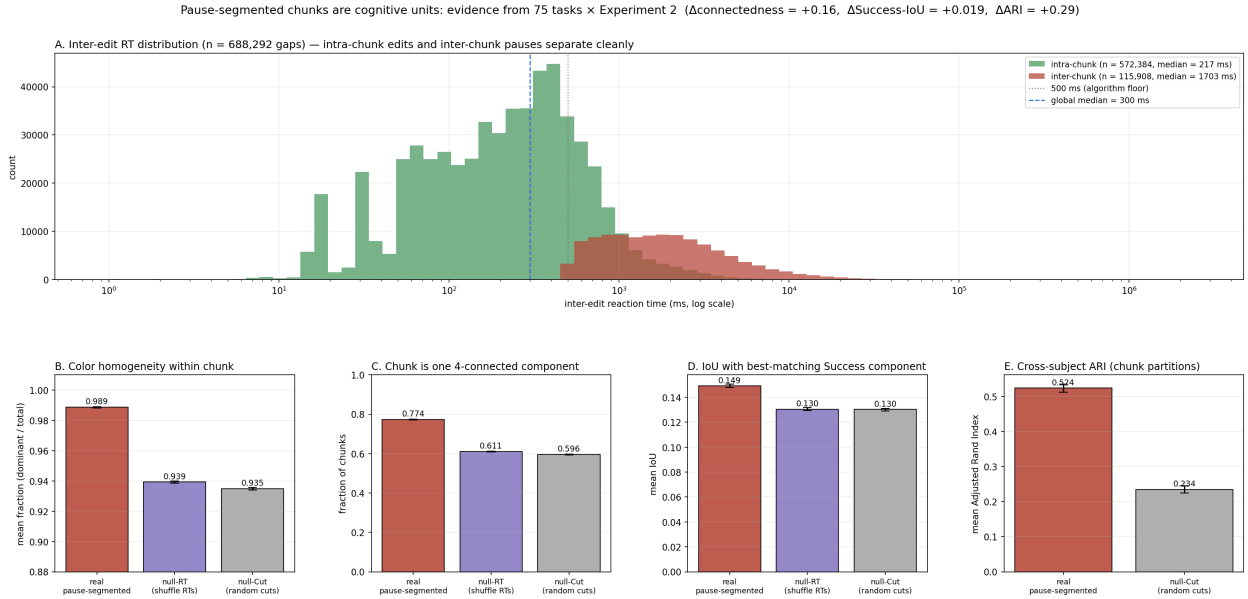

Figure 1: Pause-segmented chunks are cognitive units. **A.** Distribution of inter-edit reaction times on log-scale (n = 688,308 gaps). Intra-chunk and inter-chunk distributions separate with little overlap; the intra-chunk median is 217 ms (motor execution) while the inter-chunk median is 1,703 ms (cognitive planning)—a 7.8× gap. **B–D.** Structural coherence of chunks vs. two null re-segmentations. Real chunks have higher color homogeneity (0.989 vs. 0.935–0.939), are far more often a single connected component (0.77 vs. 0.60), and align better with Success components (0.149 vs. 0.130). Error bars are bootstrapped 95% CIs. **E.** Adjusted Rand Index between chunk partitions of random subject pairs. Real pairs agree more than twice as strongly as null-Cut pairs (0.52 vs. 0.23), showing that subjects converge on a shared task-level segmentation rather than chunking idiosyncratically.

(A) Inter-edit RTs are bimodal. Of 688,308 inter-edit gaps, the 83.2% below each subject's pause threshold have median 217 ms—consistent with pen-and-mouse motor execution and well below the 500 ms algorithmic floor. The 16.8% above threshold have median 1,703 ms, which is long enough to be cognitive planning time. The two distributions overlap little (Figure 1A), so the pause threshold is not slicing a long tail: it is identifying *a qualitatively different class of action-to-action gaps.*

(B–D) Real chunks beat both nulls on every structural metric. Real chunks are nearly colour-pure (0.989 homogeneity) vs. 0.935 for null-Cut, reflecting that subjects empty one color before reaching for the next. Most strikingly, real chunks form one 4-connected component 77.4% of the time vs. 59.6% for null-Cut and 61.1% for null-RT. Because null-Cut preserves chunk *size* distribution exactly, this +17.8 pp gap in connectedness cannot be explained by chunk length: pauses happen at spatial boundaries, not at random points within continuous drawing bursts. Chunks also match Success-grid connected components 14.5% better in IoU than chance-cut chunks, showing that cognitive-unit boundaries align with the task's underlying object structure. All three gaps are substantially larger than their bootstrapped 95% CIs.

(E) Subjects agree on how to chunk each task. If chunking were idiosyncratic, the cross-subject ARI would collapse toward the null-Cut value once chunk count and size are held fixed. Instead, real subject pairs have mean ARI = 0.52 against null-Cut = 0.23. The gap of +0.29 means most of the ARI difference between "identical" (1.0) and "chance" (0.0) is *captured* by the task's structure: different participants look at the same CogARC task and decide to draw the same things together.

Conclusion

Pause-segmented chunks pass three independent tests of being cognitive units. Inter-edit RTs decompose into a fast motor regime and a slow planning regime with little overlap (Figure 1A). Within-chunk coherence beats temporally-matched null segmentations on color, on connectedness, and on alignment with object boundaries (Figure 1B–D). And the chunking subjects arrive at is shared across the population rather than idiosyncratic (Figure 1E). Taken together, these results establish that when we say "this subject drew n chunks," we are reporting the subject's own count of how they mentally decomposed the task, not an imposed segmentation. All downstream analyses and the imitation-target for our step-by-step GNN rest on this foundation.